

WASP-43b

R-0420

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_250125 · created 2026-07-19T08:09:59+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460700.9327 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.197 +/- 0.058
Transit depth (Rp/Rs)^2	3.87 +/- 2.28 [%]
Orbital Inclination (inc)	87.0 +/- 2.83
Airmass coefficient 1 (a1)	0.057 +/- 0.035
Airmass coefficient 2 (a2)	1.33 +/- 0.14
Scatter in the residuals of the lightcurve fit is	40.35 %
Best Comparison Star	None
Optimal Aperture	2.88
Optimal Annulus	3.75
Transit Duration (day)	0.0614 +/- 0.0047

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.197 ± 0.058 vs 0.1594 (deviation 0.25σ)
Duration fit vs published	0.0614 d vs 0.0512 d (deviation 0.82σ)
Mid-time O–C	-5.9 min
Depth SNR	1.3
Residual scatter	40.35 %

Light curve

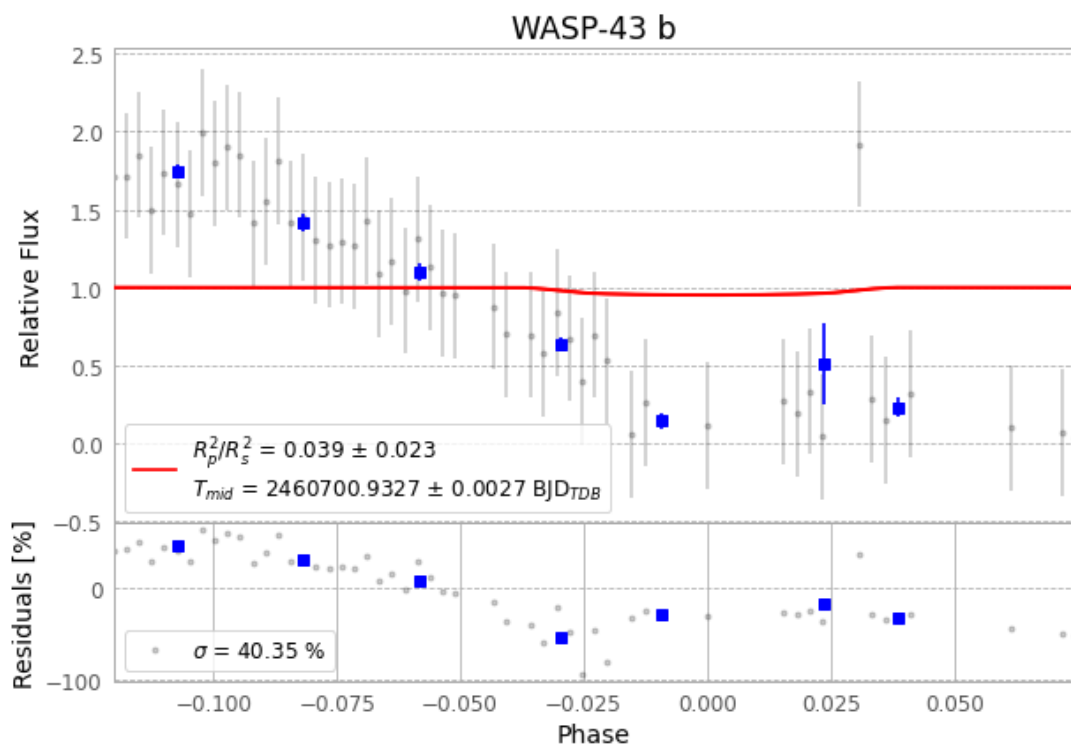


Figure 1 — FinalLightCurve_WASP-43 b_2025-01-25.png (SHA-256 05c03c4857666baa...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [283, 403], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 aafc3d1d53115d4c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

81 FITS frames (53 MB), WASP-43250125075416.FITS → WASP-43250125115415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-250125.log (b3097df242e6...), FinalLightCurve_WASP-43 b_2025-01-25.png (05c03c485766...), FinalLightCurve_WASP-43 b_2025-01-25.csv (7cd936df191e...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 08:10Z.